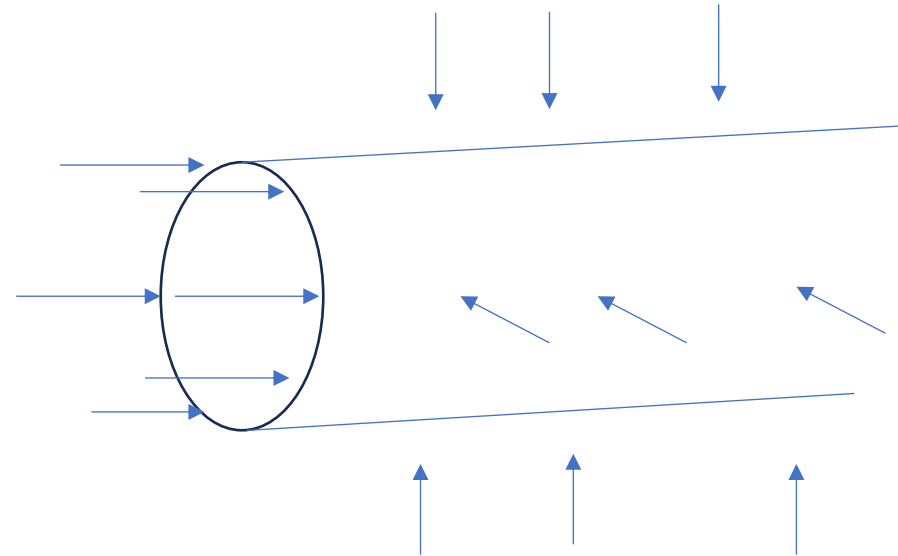


Ritz for Cylinders Under External Pressure

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$$N_z = -\frac{pR}{2} - N_A$$

$$N_\theta = -p \cdot R$$



Formulation

Ritz Method for **Stability Analysis** of Composite Cylinders

- Potential of the external load

$$V = -\frac{1}{2} \int_S N_z \left(\frac{\partial w}{\partial z} \right)^2 + N_\theta \left(\frac{\partial w}{\partial \theta} \right)^2 dS$$



$$\frac{\partial w}{\partial z} = [N_{w,z}] \{q_w\}$$

$$\frac{\partial w}{\partial \theta} = [N_{w,\theta}] \{q_w\}$$

$$V = -\frac{1}{2} \int_S N_z \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix}^T \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,z}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}^T \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix} + N_\theta \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix}^T \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,\theta}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}^T \begin{Bmatrix} \{q_u\} \\ \{q_v\} \\ \{q_w\} \\ \{q_{\beta z}\} \\ \{q_{\beta \theta}\} \end{Bmatrix} dS$$

$$W = -V = \frac{1}{2} \int_S N_z \{q\}^T [H_z]^T [H_z] \{q\} + N_\theta \{q\}^T [H_\theta]^T [H_\theta] \{q\} dS$$

$$W = \frac{1}{2} \int_S \{q\}^T \left\{ N_z [H_z]^T [H_z] + N_\theta [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

$$[H_z] = \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,z}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}$$

$$[H_\theta] = \begin{bmatrix} [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [N_{w,\theta}] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \\ [0] & [0] & [0] & [0] & [0] \end{bmatrix}$$

$$N_z = -\frac{p \cdot R}{2} - N_A$$

$$\sigma = \frac{F}{A} = \frac{P\pi R^2}{2\pi R t} = \frac{N_A}{t}$$

$$N_A = -\frac{p \cdot R}{2}$$

$$N_z = -\frac{p \cdot R}{2} - \frac{p \cdot R}{2} = -p \cdot R$$

$$N_\theta = -p \cdot R$$

$$N_\theta = N_z = \lambda$$

$$W = \frac{1}{2} \int_S \{q\}^T \left\{ N_z [H_z]^T [H_z] + N_\theta [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

$$W = \frac{\lambda}{2} \int_S \{q\}^T \left\{ [H_z]^T [H_z] + [H_\theta]^T [H_\theta] \right\} \{q\} dS$$

- In the equilibrium

$$\delta\pi_p = \delta U - \delta W = 0$$

$$\delta W = \frac{1}{2} \int_S N_z \{ \delta q \}^T [H^*]^T [H^*] \{ q \} dS + \frac{1}{2} N_z \int_S \{ q \}^T \left\{ [H_z]^T [H_z] + 2[H_\theta]^T [H_\theta] \right\} \{ q \} dS$$

$$\delta\pi = \{ \delta q \}^T \left\{ \int_S [H]^T [C^*] [H] dS + N_z \int_S \{ q \}^T \left\{ [H_z]^T [H_z] + 2[H_\theta]^T [H_\theta] \right\} \{ q \} dS \right\} \{ q \} = 0$$

$$[K_{Gz}] = \int_S \left\{ [H_z]^T [H_z] + 2[H_\theta]^T [H_\theta] \right\} dS$$

$$\{ [K] + \lambda [K_{Gz}] \} \{ q \} = \{ 0 \}$$

$$p_{cr} = -\frac{\lambda}{R}$$

Cylinder buckling mode shape

